

Crop Classification from Multi-temporal Satellite Images

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Abstract—Accurate crop type classification using remote sensing is critical for agricultural monitoring and food security assessment. This project leverages Google Earth Engine (GEE), Folium, geemap, and TensorFlow to develop and evaluate machine learning models for crop classification using satellite data with multi-temporal dimension. The objective is to compare Long Short-Term Memory (LSTM), Convolutional Neural Network (CNN), Random Forest (RF), and Support Vector Machine (SVM) which are classification models trained on the locational data generated by GEE using spectral and temporal features from Sentinel-2 imagery and USDA NASS Cropland Data Layer (CDL) labels for 2024.

I. INTRODUCTION

In this project, we leverage Google Earth Engine, Folium, geemap, and TensorFlow to develop and evaluate machine learning models for crop classification using multi-temporal satellite data. The primary objective is to compare multiple classification models such as LSTM, CNN, Random Forest, and SVM, using spectral and temporal features extracted from Sentinel-2 imagery and crop labels from USDA NASS Cropland Data Layer for the year 2024.

We define two areas of interest for the data gathering process. An easy region with four dominant crop classes, and a hard region with over twenty classes. For each area of interest, 150 labeled points per class are sampled. Features including bands B2, B3, B4, and B8, along with Normalized Difference Vegetation Index (NDVI) derived from B4 and B8, are extracted. Cloud coverage is accounted for during data retrieval. The resulting datasets are exported as Excel files and preprocessed to filter out irrelevant land cover types, remove duplicated data and remove points with insufficient amounts of time stamps.

Data is processed in both mono-temporal and multi-temporal formats. In the multi-temporal setting, points with fewer than ten timestamps are excluded, and duplicate spatial-temporal entries are removed. The duplication removal is the same for the mono-temporal format. Summary statistics such as date range, per-point timestamp counts, and observation frequency are also computed.

Visualization includes spatial distribution of samples, class distribution, timestamp statistics, and NDVI time series (for multi-temporal data). Among the models, LSTM achieves the highest accuracy across all configurations. RF offers the best trade-off in speed and performance followed closely by SVM, particularly for baseline mono-temporal data.

II. PROBLEM DESCRIPTION

Effective and scalable crop type classification is a critical component of modern agricultural monitoring systems. Accurate identification of crop types over large areas supports a wide range of applications, including yield estimation, land use management, market forecasting, food security analysis, and environmental impact assessment. Traditionally, such classification has relied on manual surveys and local field data, which are often expensive, time-consuming, and difficult to scale. In recent years, remote sensing combined with machine learning has emerged as a alternative, enabling large-scale, automated classification using satellite imagery.

Despite the progress in remote sensing-based classification, several challenges persist. These include high intra-class variability due to different growing stages of the same crop differing in looks from each other (e.g. a growing wheat crop may look brown, green or yellow depending on the growth season), inter-class similarity due to similarities between crops (e.g. wheat and barley grow in similar growth periods and have similar colours), cloud contamination in satellite imagery, and the need for high-quality labeled data. Lastly, using time-based data that shows changes in vegetation, especially for crops with clear growth stages, makes designing and training the model more complicated.

This project addresses the following core research question:

How do different machine learning models perform in classifying crop types using mono-temporal versus multi-temporal satellite imagery, and how does classification complexity affect model accuracy and efficiency?

To explore this question, we formulate a pipeline that uses publicly available satellite data and machine learning techniques to classify crop types in two distinct agricultural regions in the U.S. using CDL as a source of ground truth labels, and Sentinel-2 data to extract spectral features.

Our classification scenarios include:

- **Easy** classification task with four dominant crop classes in eastern Arkansas.
- **Hard** classification task with over twenty crop classes in eastern North Carolina.

Each classification task is conducted under two data configurations:

- **Mono-temporal**, where a single observation per point is used.

- **Multi-temporal**, where multiple observations per point are used over the growing season.

We employ four machine learning models for crop land classification which are LSTM, CNN, RF, and SVM and compare their performance in each configuration. We aim to evaluate the impact of temporal data on classification performance, the relationship between classification complexity and model performance and the trade-offs between model accuracy and computational efficiency.

III. METHODS

This project implements a comprehensive workflow for crop type classification using satellite imagery and machine learning. The process involves several key stages: data collection, preprocessing, dataset structuring, model training, and evaluation.

A. Study Areas

We defined two Areas of Interest (AOIs) to simulate classification tasks of varying complexity:

- **Easy AOI (Eastern Arkansas):** A region with lower crop diversity ($n = 4$) [-90.3313, 35.5684, -90.2865, 35.5983].
- **Hard AOI (Eastern North Carolina):** A region with higher crop diversity ($n > 20$), [-77.5348, 35.9244, -77.7215, 36.0472]

B. Label Collection

Ground truth labels were extracted from the 2024 USDA NASS Cropland Data Layer, an annually updated, crop-specific land cover dataset for the continental United States. The CDL is generated by the National Agricultural Statistics Service (NASS), Research and Development Division, Geospatial Information Branch, Spatial Analysis Research Section. The CDL provides class labels as numeric codes along with associated color values for visualization. Each code corresponds to a specific crop type, with examples including Corn (1, #ffd400), Cotton (2, #ff2626), Rice (3, #00a9e6), Soybeans (5, #267300), and Winter Wheat (24, #a87000), among many others. For each Area of Interest, we randomly sampled 150 labeled points per crop class. Classes considered irrelevant or likely to introduce noise—such as open water, urban/developed areas, barren land, or fallow ground, were excluded from sampling. The numerical CDL codes were mapped to human-readable crop names for better readability during pre-processing. This mapping also included the standard CDL color codes to enable consistent labeling in visualization outputs, such as maps and class distribution charts.

C. Feature Extraction

Spectral and temporal features were extracted from Copernicus Sentinel-2 Harmonized Surface Reflectance (H-SR) imagery available through Google Earth Engine. Sentinel-2 is a high-resolution, multi-spectral imaging mission developed by the European Space Agency (ESA) as part of the Copernicus Program. It supports applications in agriculture, forestry,

water resources, and land monitoring. From each image, we extracted the following Level-2A spectral bands:

- **B2 (Blue)** — 10m resolution, 492–497 nm
- **B3 (Green)** — 10m resolution, 559–560 nm
- **B4 (Red)** — 10m resolution, 664–665 nm
- **B8 (Near Infrared, NIR)** — 10m resolution, 833–835 nm

Using the red (B4) and NIR (B8) bands, we computed the Normalized Difference Vegetation Index (NDVI) at each timestamp as:

$$NDVI = \frac{B8 - B4}{B8 + B4}$$

NDVI shows how healthy and active vegetation is over time and was central to the temporal modeling strategies applied.

To ensure data quality, only cloud-free observations were retained. GEE's built-in cloud masking approach was applied using the QA60 band. Future studies may incorporate Red Edge or SWIR bands to enrich the spectral feature space, particularly for distinguishing subtle crop differences or stress responses.

D. Preprocessing Dataset Construction

Two dataset formats were prepared for each AOI:

- 1) **Mono-temporal:** A single observation per point from the middle of the growing season was used to ensure crop differentiation.
 - Removing duplicate entries with identical (lat, lon, date) keys.
 - Reporting per-point statistics.
- 2) **Multi-temporal:** Full spectral time series were collected across the season. Quality control involved:
 - Discarding points with fewer than 10 valid timestamps.
 - Removing duplicate entries with identical (lat, lon, date) keys.
 - Reporting metadata including timestamp ranges and per-point statistics.

The cleaned datasets were exported as .csv files.

Prior to modeling, several preprocessing steps were performed to ensure data consistency, improve model performance, and prepare the dataset for both traditional and sequence-based learning algorithms. The original ground truth labels obtained from the CDL were encoded as integer class IDs. These numeric codes were mapped to human-readable crop names using a predefined lookup dictionary for better interpretability. Subsequently, a filtering step was applied to retain only agriculturally relevant land cover classes. Non-cropland classes such as Open Water, Developed Land and Fallow Ground were excluded from the dataset. This step avoided learning really small classes and ensured that the models focused solely on crop-type classification rather than broader land use classification. To standardize the feature scales across inputs and reduce the risk of model bias toward high-magnitude features, spectral bands (B2, B3, B4, B8) and NDVI values were normalized which is particularly important

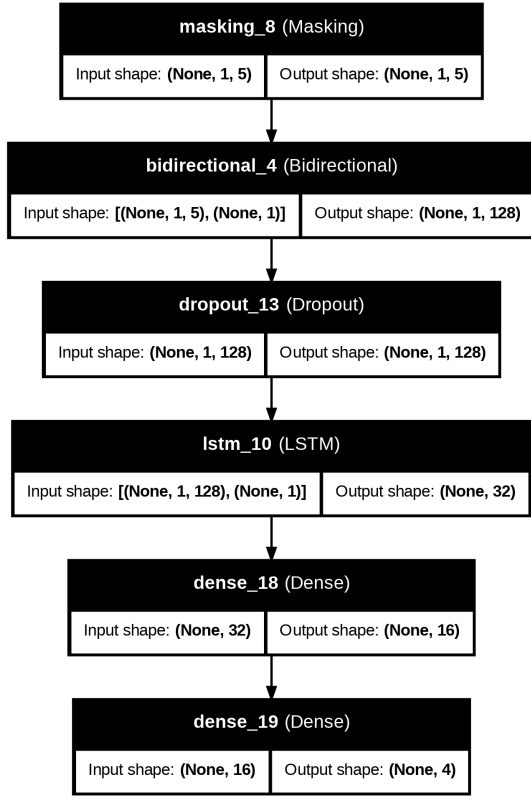


Fig. 1. LSTM architecture used for multi-temporal crop classification. Input features per timestep included NDVI and four spectral bands.

for neural network models, which are sensitive to feature scale and distribution. Satellite imagery is collected on irregular dates due to cloud cover, satellite orbit, and data availability, the resulting time series varied in length across locations. To enable batch processing and compatibility with fixed-length sequence models, all time series were aligned to a common temporal window. This was achieved by padding to the maximum amount of timestamps in a given dataset.

E. Machine Learning Models

We evaluated four machine learning models for crop classification across both the easy and hard datasets, under mono-temporal and multi-temporal configurations. Each model was selected for its unique strength in handling different data modalities:

1) Long Short-Term Memory (LSTM):

LSTM networks are a type of recurrent neural network (RNN) specifically designed to capture long-term dependencies in sequential data. In our pipeline, LSTM was applied to both mono-temporal and multi-temporal datasets to leverage the phenological patterns captured across time. Although in the case of mono-temporal it was heavily modified to accept non-sequential data as such the performance was not good.

2) Convolutional Neural Network (CNN):

CNNs are effective at extracting local patterns in struc-

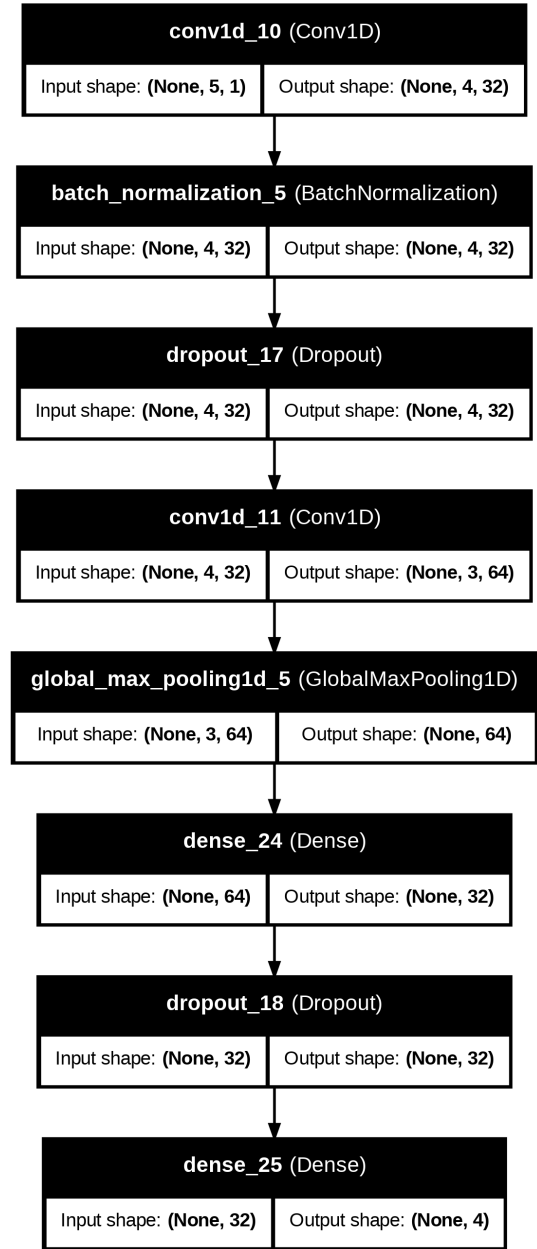


Fig. 2. CNN architecture for time-feature grid input. Applied to both mono- and multi-temporal data using 1D or 2D convolutions.

tured input such as images or time series. For mono-temporal data, we applied a simple 1D CNN directly to feature vectors. For multi-temporal data, time series were reshaped into a 2D format allowing the model to apply 2D convolutions across both time and spectral dimensions. This made CNNs capable of learning both temporal trends and inter-band relationships.

3) Random Forest (RF):

Random Forest is an ensemble method based on decision trees. It constructs multiple trees during training and outputs the mode of the classes for classification tasks. RFs are robust to overfitting, easy to implement, and require

minimal hyperparameter tuning. They were particularly effective for mono-temporal classification due to their ability to handle high-dimensional tabular data.

4) Support Vector Machine (SVM):

SVMs find the optimal hyperplane that separates classes in high-dimensional space. We used the radial basis function kernel, which is well-suited for nonlinear classification tasks. While SVMs performed well on simpler, mono-temporal, low-class-count, datasets their scalability limited their utility on multi-temporal or high-class-count datasets.

Models were implemented using TensorFlow and Scikit-learn. An 80/20 train-test split was used.

F. Visualization

Folium and geemap were used to create interactive maps and exploratory visualizations. Visual diagnostics were essential for understanding dataset characteristics and interpreting model behavior. Below are examples of key visualizations for both mono- and multi-temporal dataset configurations for both easy and hard datasets.

1. *Spatial Distribution of Labeled Points with NDVI Basemaps*: Crops are evenly distributed within the given AOI boundaries in both datasets, as shown in Figure 3. However, both the easy and hard regions demonstrate class imbalance, where certain crops dominate the label distribution. The NDVI basemaps further confirm that labeled points span various vegetation zones, providing representative coverage of land cover conditions.

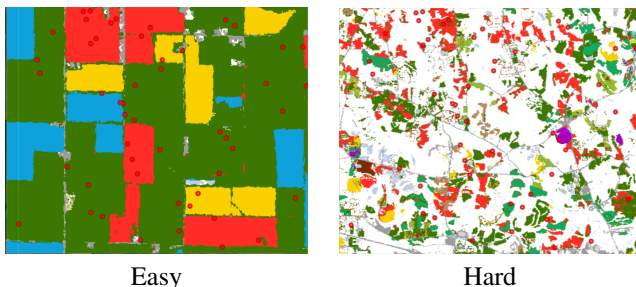


Fig. 3. Spatial distribution of labeled points over NDVI basemaps.

2. *Spatial Distribution of Labeled Points with Respect to Class*: As visualized in Figure 4, class-specific spatial distributions reveal how certain crop types cluster geographically. While crops are generally dispersed across the AOI, dominant classes such as corn or soybean tend to concentrate in specific subregions. This spatial bias may impact model generalization if some crops are limited to narrow zones.

3. *Class Distribution Histograms*: Figure 5 shows the frequency of labeled samples for each class. Both easy and hard datasets exhibit significant class imbalance, where crops like corn and soybean dominate while classes such as barley and oats are underrepresented. This imbalance is more pronounced in the hard dataset and directly affects classification performance on minority classes.

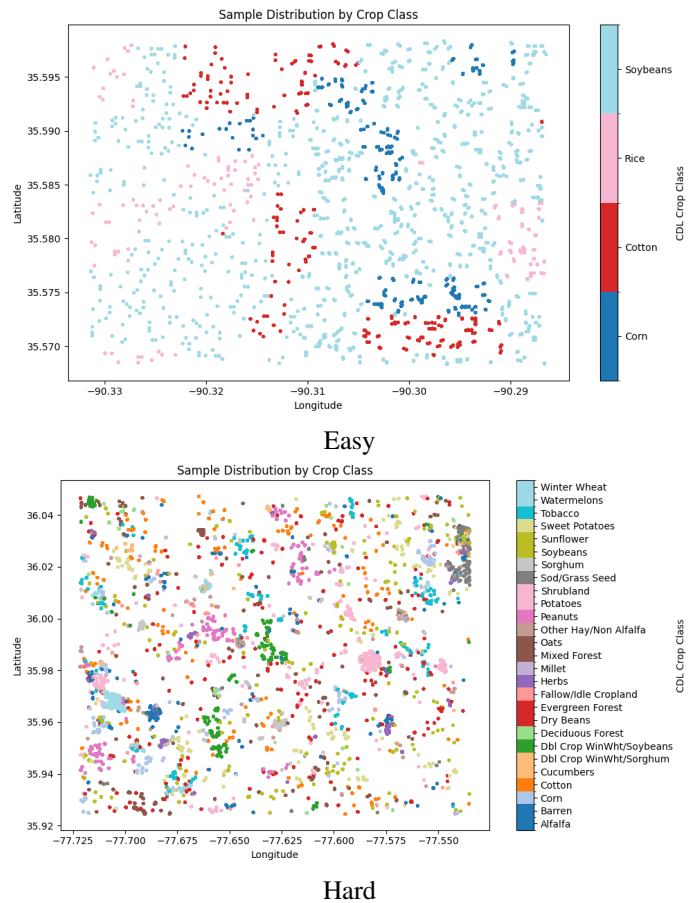


Fig. 4. Spatial distribution of labeled points by class.

4. *Distribution of Timestamps (Only Multi-Temporal)*: The distribution of valid timestamps across samples is illustrated in Figure 6. Both datasets cover observations throughout the growing season, providing sufficient temporal coverage to detect phenological patterns. Most samples include at least 10 timestamps, satisfying the minimum threshold for LSTM-based models. Overall seasonal progression is well-represented.

IV. RESULTS

To evaluate the effectiveness of our classification pipeline, we conducted a series of experiments across the aforementioned two locations (easy and hard) with two data types (mono-temporal and multi-temporal). Each of the four configurations were used to train and test four models: LSTM, CNN, RF, and SVM. The results below summarize classification performance, temporal effects, and class-level behavior.

A. Experiment Objectives

Each experiment aimed to answer the following research questions:

- 1) How does classification accuracy vary between mono-temporal and multi-temporal datasets?
- 2) Which model performs best in each setting (easy vs. hard, mono vs. multi)?

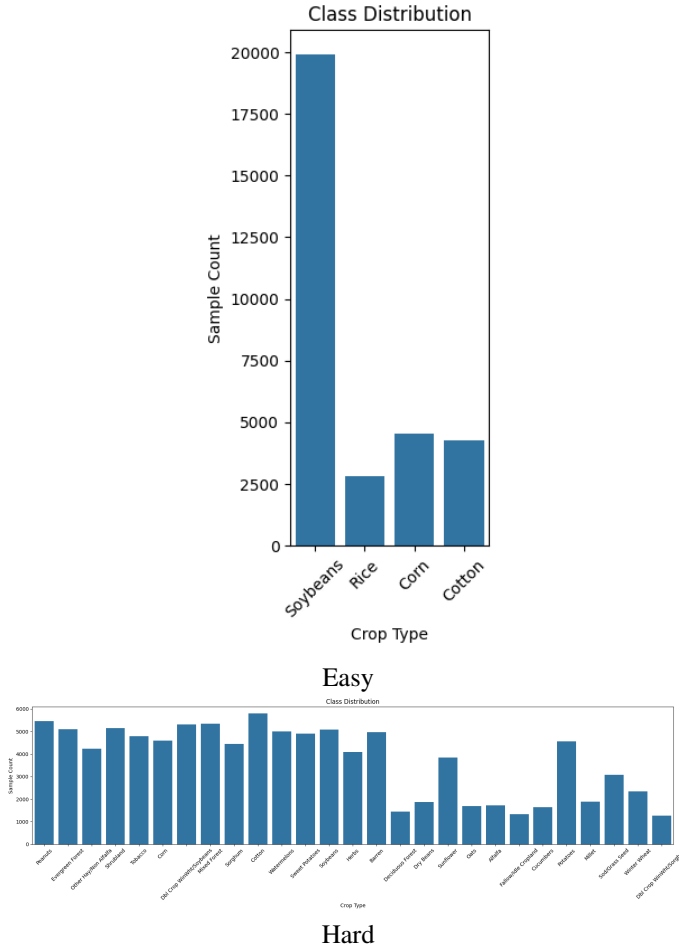


Fig. 5. Class distribution histograms across dataset configurations.

- 3) What is the impact of temporal information on class separability?
- 4) How does model performance trade off with computational cost?

B. Overall Performance

The table below summarizes classification accuracy across all 16 experiments:

TABLE I
MODEL ACCURACY ACROSS DATASET CONFIGURATIONS

Model	Easy Mono	Easy Multi	Hard Mono	Hard Multi
LSTM	0.68	0.97	0.17	0.68
CNN	0.78	0.93	0.17	0.62
RF	0.85	0.87	0.27	0.54
SVM	0.71	0.84	0.19	0.59

Key findings:

- **LSTM** demonstrated better performance on multi-temporal datasets, indicating the value of sequential NDVI information.
- **RF** performed best on mono-temporal setups, especially in the easy dataset.

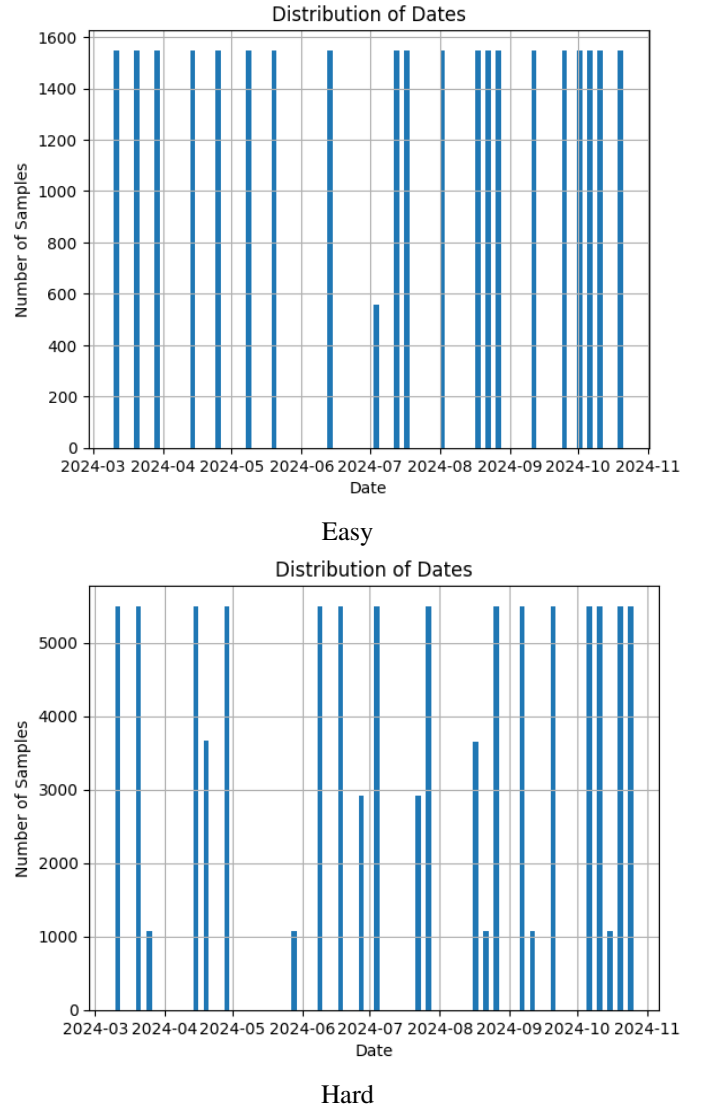


Fig. 6. Distribution of timestamps per point in multi-temporal datasets.

- **CNN** provided balanced and stable results across all cases.
- **SVM** had similar scores compared to the RF although in some instances RF was better at representing the dataset.

C. Confusion Matrices for Best Models

Below are the confusion matrices for the best-performing model in each case:

D. Class-Level Accuracy Histograms

The following histograms illustrate per class accuracy for the same four best-performing models:

E. Impact of Data Characteristics

- Strong class imbalance in the hard datasets reduced model generalizability.
- The minimum threshold of 10 valid timestamps effectively filtered low-quality samples for LSTM.

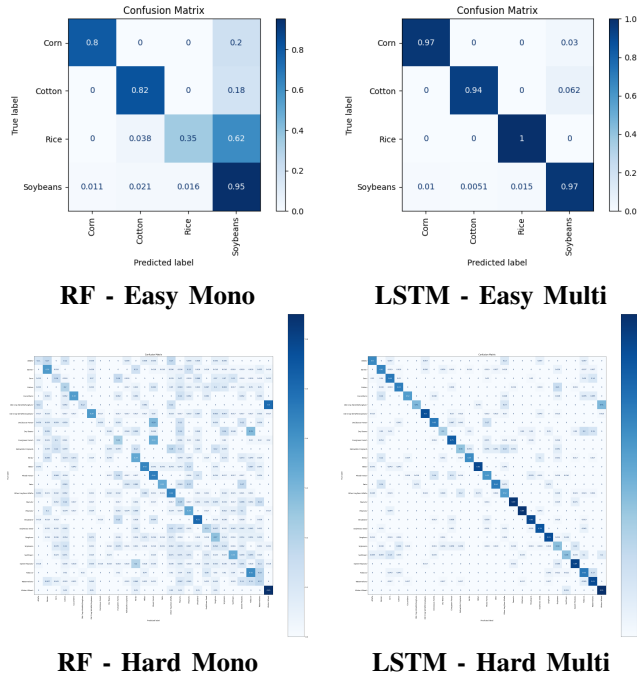


Fig. 7. Confusion matrices for best models across dataset configurations.

- Temporal signals in NDVI notably improved separation of crops with similar reflectance patterns.

V. DISCUSSION

The experimental results provide several important insights into the effectiveness of different machine learning approaches for crop classification using remote sensing data. The comparison across datasets of varying complexity and temporal configurations reveals key trends in model performance, data utility, and computational trade-offs.

A. Impact of Temporal Information

One of the most significant findings is the consistent improvement in classification accuracy when multi-temporal data is utilized. The LSTM model showed the highest performance on both the easy and hard datasets in the multi-temporal setting. This is likely due to its ability to capture phenological patterns over time, which are crucial for distinguishing crops with similar spectral signatures but differing growth cycles.

CNNs also benefited from temporal data when reshaped into 2D structures (e.g., time versus features), though not to the same extent as LSTMs. Traditional models like RF and SVM performed best on mono-temporal data and did not scale as effectively with the added temporal dimension. However with multi-temporal data, in the case of "Hard Multi" Their scores have improved to be more comparable to the LSTM and CNN. This is likely due to the fact that with more classes there is a need for more data to differentiate between crop types as mono-temporal data might not be enough for over 20 classes of crop types.

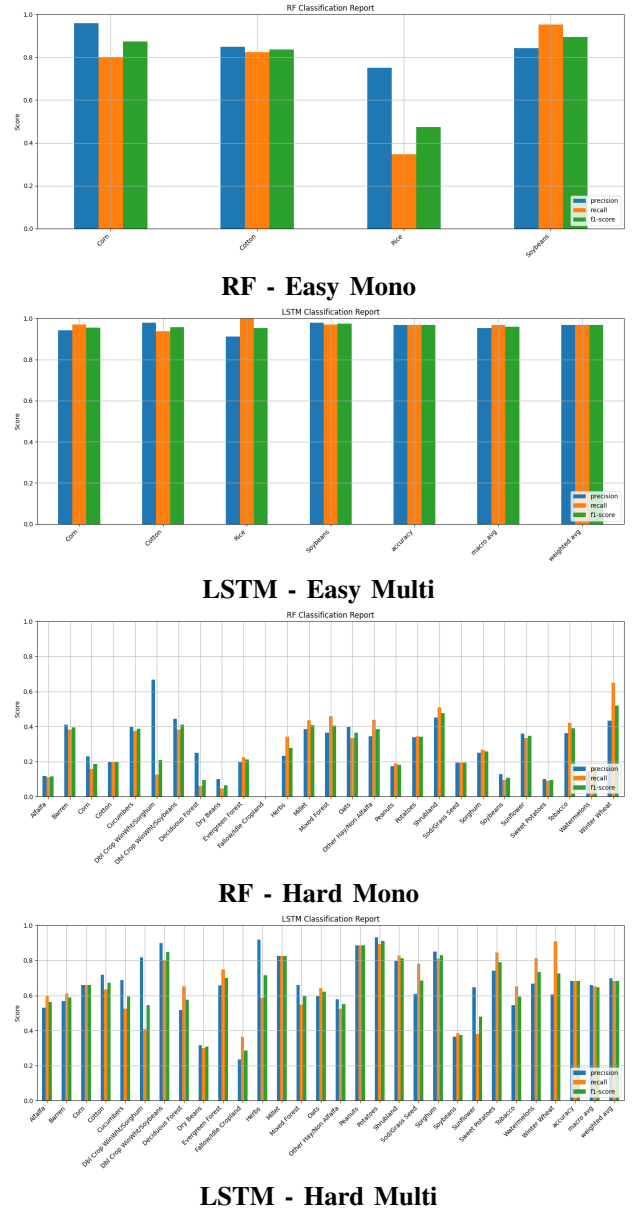


Fig. 8. Per-class accuracy histograms for top-performing models.

B. Effect of Dataset Complexity

Model performance declined across all methods as dataset complexity increased. In the hard dataset, which included more than 20 crop types, overall accuracy and F1-scores were lower than in the easy dataset (with four dominant classes). This can be attributed to:

- Increased class imbalance, leading to overfitting on dominant crops.
- Greater spectral and temporal overlap between crop types.
- Limited training samples per class, even with balanced sampling efforts.

While LSTM still achieved the highest performance on the hard multi-temporal dataset, confusion matrices revealed difficulties in distinguishing minority classes. This highlights the

need for additional training and data to differentiate between the crop types.

C. Trade-offs Between Accuracy and Efficiency

In terms of computational efficiency, RF consistently trained the fastest, followed by SVM. These models are suitable for quick baselines or deployments in resource-limited settings. However, they offer limited performance gains with multi-temporal data.

In contrast, the neural models (CNN and LSTM) required more training time and tuning but achieved higher accuracy, especially in scenarios with rich temporal information. These trade-offs suggest that model selection should be done on a case-by-case basis where the cost, speed and accuracy are assessed together to fulfill a specific goal.

D. Limitations

While promising, the current study has several limitations:

- The use of 150 samples per class may be insufficient for rare crops, especially in high-diversity datasets.
- Cloud masking, while robust, may still admit artifacts that degrade temporal signal quality.
- Mono-temporal data relied on a single mid-season date, which may not represent all crop phenologies equally.

Future work could explore enhanced sampling strategies, working with more data to search the limit for accuracy for the current configurations, and using more advanced sequence models such as Transformers to better leverage temporal dependencies.

VI. CONCLUSIONS

This study demonstrates the feasibility of combining remote sensing and machine learning techniques for crop type classification across spatial and temporal variations. Leveraging Google Earth Engine, Sentinel-2 imagery, and USDA Cropland Data Layer labels, we developed and evaluated a reproducible pipeline capable of classifying crops in two distinct agricultural regions using both mono-temporal and multi-temporal datasets.

Our results highlight the significant benefits of temporal information. Models such as LSTM, which are designed to capture sequential patterns, achieved the highest classification accuracy on multi-temporal data. Among the four evaluated models: LSTM, CNN, RF, and SVM, LSTM consistently outperformed the others in temporal scenarios, while RF provided the most efficient performance in terms of training time and computational requirements.

We also found that dataset complexity substantially influenced model accuracy. Careful dataset construction and pre-processing, including filtering and relabeling, proved critical to success. Visualizations such as NDVI time series and spatial class distributions were quite useful in diagnosing model behavior and informing methodological decisions.

Future directions for this research include:

- Exploring enhanced sampling strategies to improve data quality and coverage.

- Using larger datasets to test the accuracy limits of the current model configurations.
- Investigating advanced sequence models, such as Transformers, to better capture temporal dependencies.

APPENDIX

Author Contribution

This project was completed by Sezer Enis Birgili the only person on the project as well as the author of this report. All stages of the research—data collection, preprocessing, modeling, analysis, and visualization—were carried out solely by the author.

Additional Results

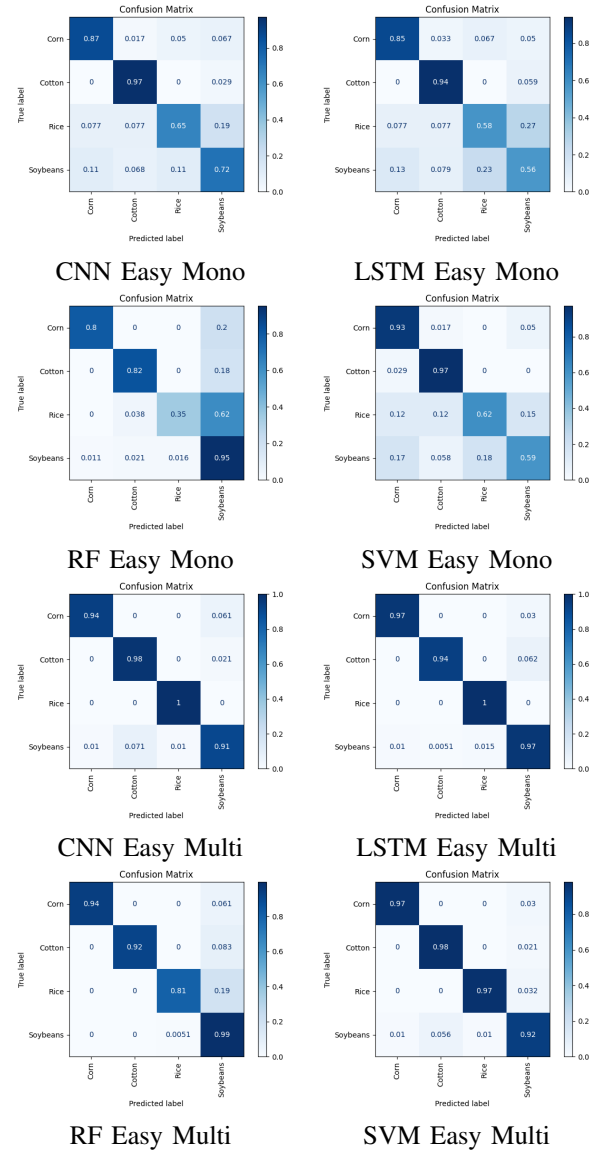


Fig. 9. All confusion matrices for each model and dataset configuration.

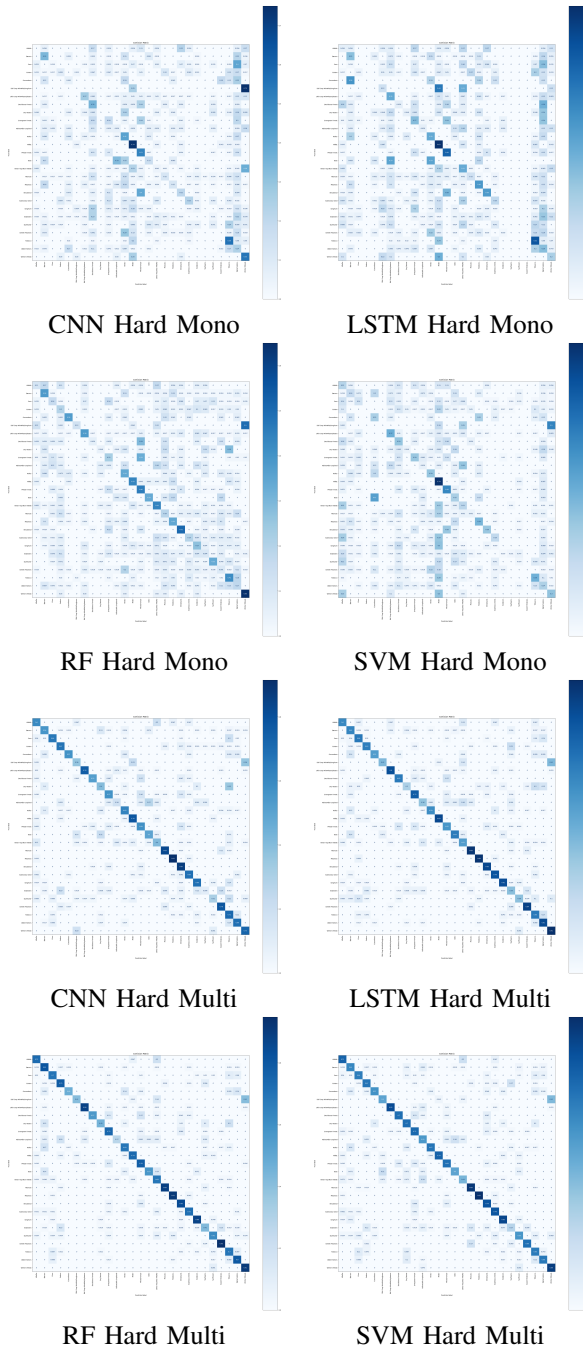


Fig. 10. All confusion matrices for each model and dataset configuration.

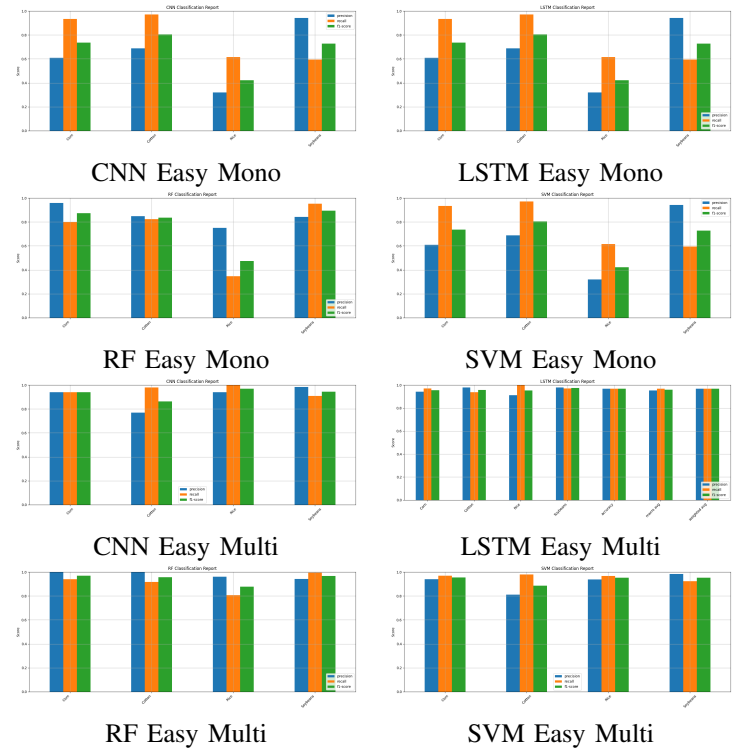


Fig. 11. Per-class accuracy metrics (precision, recall, F1) across all dataset configurations and models.

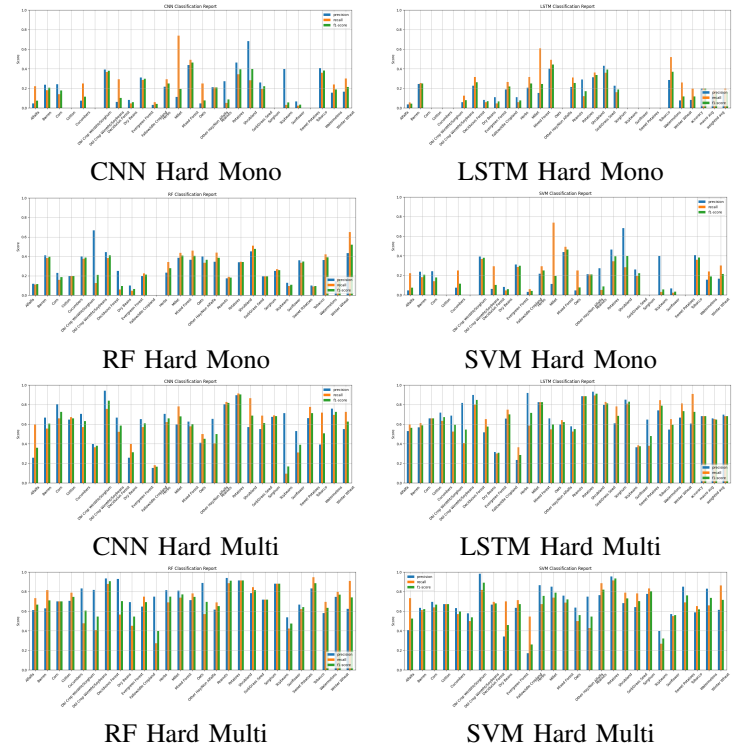


Fig. 12. Per-class accuracy metrics (precision, recall, F1) across all dataset configurations and models.